nPnB GREEX: Graph Entities Explorer
Quick User Guide

1 Graph Exploration

Networks are one of the main conceptual structures used to model complex systems: nodes represent the lowest-level elements of the system, while edges represent their interactions. Graph clustering algorithms are then used to identify over-densely connected sets of nodes representing higher-level Entities of the system. **bec** is a graph clustering algorithm that identifies the **Entities** in a network according to a **description scale** within the **nPnB** framework.

> **nPnB:** B. Gaume, I. Achitouv, and D. Chavalarias: *Two antagonistic objectives for one multi-scale graph clustering framework.* Nature Scientific Reports, 15 (2025), p. 13368. `https://hal.science/hal-05046050v1`
>
> **bec:** B. Gaume: *Finding optimal clusterings in the nPnB framework is hard. BEC.2: an algorithm to find high-performing ones.* Forthcoming (2026).
>
> **Description Scales:** B. Gaume: *Defining the appropriate scales for describing a complex system.* Forthcoming (2026).

To visualize graphs, we use the 3D Force-Directed Graph Layout developed by Vasco Asturiano, where the attractive forces of links build the graph layout. `https://github.com/vasturiano`

2 Gbase

The input is called **'Gbase'**; it is an undirected graph $G_{base} = (V_{base}, E_{base})$ whose vertices are grouped into modules representing Entities at different scales of description. The vertices of this graph are called **'Basic Nodes' (BN)**.

3 Graph of Scale Entities

In the main window, you can choose a **Scale of Description** s to explore Gbase. The Basic Nodes of Gbase are then grouped into Entities that form a partition of the BNs at scale s (The corresponding partitional clustering score is displayed below). One can then explore the graph of Gbase Entities at scale s: Vertices represent Entities of Gbase at scale s, and two Entities X_i and X_j are connected $\iff \exists x_i \in X_i$, $\exists x_j \in X_j$ such that $\{x_i, x_j\} \in E_{base}$.

3.1 Mouse Actions

Clicking: Vertex-Entity/Background
Clicking a Vertex-Entity applies actions related to the selected Entity. Clicking on Background applies global actions on Entities.

3.1.1 Clicking Vertex-Entity

Left Click	Zoom on the selected Entity.
Right Click	(⊡) → [⊡]: (This ENTITY) → [Visible ENTITIES] Open a visibility panel centered on the selected Entity. Typical operations include showing or hiding neighbouring Entities.
Ctrl + Left Click	Toggle Entity label visibility for the selected Entity.
Shift + Left Click	Toggle label-Entity visibility of neighbors Entity when mouse hover.
Ctrl + Right Click	Displays the list of BNs belonging to the selected entity.

3.1.2 Clicking Background

Left Click	No action. Useful for clicking on empty space.
Double Left Click	Toggle graph rotation.
Right Click	(⊡ ⊡ ⊡) → [⊡]: (ENTITY Properties) → [Visible ENTITIES] Open the global visibility panel. Typical operations include Entity size filtering, Entity degree filtering, label-Entity regular-expression filtering.
Shift + Left Click	Place all visible Entities in the center of the scene.
Ctrl + Left Click	Hide labels-Entities on all currently visible Entities.

3.2 Toolbar

⚙	Open the Preferences panel
❄	Enable or disable Vertex-Entity dragging.
⥁	Toggle graph rotation (+ Increase speed, - Decrease speed)
🔍	Open the FOCUS panel to: Find an Entity containing a Basic Node; (●) → [⊡]: (This Basic Node) → [Visible ENTITIES]; (● ● ●) → [⊡]: (Basic Node properties) → [Visible ENTITIES].
SubGraphBNC	Explore in a new window, the SubGraph of the Basic Nodes Covered by the current visible Entities.
UserGuide	Quick User Guide
Export	Export the current graph view as an image.

4 SubGraph of Basic Nodes

We clicked the [SubGraphBNC] button in the main window, and the secondary window opened to explore the SubGraph of Basic Nodes Covered by the Entities currently visible in the main window.

4.1 Description Scale, Color and Layout

• Two BNs sharing the same color belong to the same Entity at the active Description Scale s in the secondary window. The **[FrozCol]** button (Frozen colors) allows you to freeze node colors at the active Description Scale s in the secondary window.

• The **[ScaleAware]** button (Scale Aware) allows you to weaken the attractive force of links whose endpoints do not belong to the same Entity at the active Description Scale s in the secondary window. This modifies the Graph Layout according to the active Description Scale s.

• Combining **[FrozCol]** at a Description Scale s_1 with **[ScaleAware]** at another Description Scale s_2 allows simultaneous observation of the SubGraph at two different Description Scales. This often reveals the fractal organization present in many real-world graphs.

4.2 Mouse Actions

Clicking: Vertex-BN/Background
Clicking a Vertex-BN applies actions from the selected BN.
Clicking on Background applies global actions on BNs of the SubGraph.

4.2.1 Clicking Vertex-BN

Left Click	Zoom on the selected BN.
Right Click	Open a label-BN visibility panel centered on the selected BN. Typical operations include showing : -Me & My Neighborhood labels in the SubGraph; -and/or labels of BNs of the SubGraph within my entity, at the active Description Scale of the secondary window.
Ctrl + **Left Click**	Toggle BN label visibility for the selected BN.zzzz
Shift + **Left Click**	Toggle label-BN visibility of neighbors BN when mouse hover.

4.2.2 Clicking Background

Left Click	No action. Useful for clicking on empty space.
Shift + **Left Click**	Place all BNs of the SubGraph in the center of the scene.
Double Left Click	Toggle graph rotation.
Ctrl + **Left Click**	Hide all labels-BNs of the SubGraph.

4.3 Toolbar

❃	Toggle Vertex-BN dragging.
⥁	Toggle graph rotation (+ Increase speed, - Decrease speed)
🔎	Open the FOCUS panel to: Find a Basic Node in the SubGraph;
ScaleAware	Modify the Graph Layout according to the active Description Scale of the secondary window. Links connecting BNs belonging to different Entities are progressively weakened. (+ weakens more, - weakens less)
FrozCol	Freeze node colors at the active Description Scale of the secondary window.
UserGuide	Quick User Guide
Export	Export the current graph view as an image.

5 Tips for Exploring Large Graphs

(1) In the main window, select a Description Scale;

(2) Adjust entity visibility while keeping a close eye on the number of Basic Nodes Covered by these entities (indicated in the State (BNC));

(3) When the number of Basic Nodes Covered by the visible Entities seems reasonable, Click [SubGraphBNC] to explore, in a secondary window, the Gbase SubGraph generated by the Basic Nodes Covered.